E-Commerce Dataset: Products, Customers, and Trends

Final Project Report – **DA 503 Data Visualization with Tableau**
Student: **Ch S Ch N Venkata Satya Sai Krishna Grandhi** (ID: 5225881)
Instructor: **Michael Graham-Cornell**

1. Introduction

E-commerce platforms continue to grow as a dominant retail channel, offering customers a wide assortment of product choices, faster shipping, and increasingly competitive pricing. With this expansion, companies face challenges in understanding what drives revenue, identifying high-risk product categories, and balancing customer demand with operational efficiency. Data visualization plays a key role in revealing such patterns quickly and clearly.

This project applies Tableau to analyze a multi-category e-commerce dataset and answer strategic business questions including:

- Which customer segments and regions generate the most revenue?
- Which categories and product types are high risk due to returns or quality issues?
- How do price, discounts, popularity, and shipping choices influence product performance?
- How do operational patterns such as shipping method usage impact cost and efficiency?

To address these questions, I developed **twelve worksheets** and combined them into **two interactive dashboards**:

- **Dashboard 1 — "Global Revenue & Customer Insights"**: Provides a customer-centric view of revenue patterns, demographics, and geographic contribution.
- **Dashboard 2 — "Product Performance & Risk Intelligence"**: Focuses on product-level insights including returns, pricing behavior, popularity, and shipping efficiency.

Both dashboards incorporate highlight and filter actions, allowing users to click any data point and dynamically update the rest of the visualization. The result is a complete and interactive analytical story without requiring coding or advanced statistical modeling.

2. Data

The project uses a curated "Diversified E-Commerce Dataset" consisting of **1,000+ product-level purchase records** across multiple categories. Each row represents an individual product transaction with pricing, customer demographic, product behavior, and operational attributes.

2.1 Key Dimensions

- **Product Name** – represents product type (e.g., Running Shoes, Bluetooth Speaker)
- **Category** – Apparel, Electronics, Home Decor, Beauty, Sports
- **Customer Gender** – Male, Female
- **Customer Age Group** – 18–25, 26–35, 36–45, 46+
- **Country / City** – customer location
- **Shipping Method** – Standard, Express, Economy
- **Seasonality Flag** – "Seasonal" or "Non-Seasonal"

2.2 Key Measures

- **Price** – original list price
- **Final Price** – price after discount
- **Discount** – percentage discount applied
- **Popularity Index** – customer engagement score
- **Return Rate** – % likelihood of return
- **Return Severity** – low/medium/high risk
- **Shipping Cost** – cost associated with the shipping method
- **Revenue** – final price used for financial contribution analysis

2.3 Data Cleaning & Preparation

The dataset was imported directly into Tableau, requiring only light cleaning:

- Verifying numeric and categorical field types
- Formatting monetary fields
- Creating derived fields such as **Age Group**, **Revenue**, and **Customer Value Band**
- Splitting Country into usable geographic components for mapping

The dataset did not require heavy preprocessing, enabling the focus to remain on clear and effective visualization.

3. Related Work

Public datasets play a major role in helping students and analysts explore real e-commerce behavior. The dataset used in this project comes from Kaggle and includes product details, pricing, customer information, discounts, returns, and shipping attributes [1]. Because it combines customer, product, and operational fields in one place, it allows for a wide range of analyses, from revenue patterns to product risk and shipping performance.

Other e-commerce datasets on Kaggle offer similar structures and are often used for sales trend analysis, customer behavior studies, and transaction-level exploration [2]. These public datasets have become a common foundation for learning analytics and building dashboards.

In recent work, researchers have shown that data dashboards help businesses understand customer demand, pricing patterns, and return behavior more effectively. Studies also highlight the value of visual analytics for supporting decisions in marketing, product management, and operations [3], [4]. These findings support the approach taken in this project: using interactive dashboards to explore revenue drivers, product performance, and operational issues in a simple and accessible way.

4. Methods

I designed each worksheet to answer one focused business question, then combined them into two dashboards: one centered on customer and revenue behavior, and one focused on product-level performance, pricing, returns, and shipping patterns.

4.1 Key Worksheets

- **Revenue by Country** – Map showing the revenue of different countries.
- **Revenue by Category - Bubble Chart** – A bubble chart showing the revenue of each category

- **Revenue by Product Tree Map across Category** – A tree map showing the revenue of each product across different categories.
- **Revenue by Age Group** – A heatmap showing the revenue by age group across each category
- **Gender Pie Chart** – A Pie chart showing the revenue and count cross each gender.
- **Product Portfolio Overview** - A bar plot showing how many product types exist, helping explain catalog size and mix.
- **Return Severity by Product Name** - A bar chart ranking product types by average return severity, highlighting which items are most problematic.
- **Final Price Ranking (Lollipop Chart)** - A dual-axis lollipop chart that ranks product types from lowest to highest final price, revealing premium vs. budget segments.
- **Discount Profile by Product Name** - A bar chart showing average discount percentage per product type, indicating where markdowns are most common.
- **Popularity (Product Bubbles)** - A packed-bubble visualization where size and color reflect Popularity Index, making it easy to see highly demanded products.
- **Shipping Method × Product Heatmap** - A heatmap linking product types to shipping methods, colored by average shipping cost to show logistical patterns.

4.2 Dashboard Design

Dashboard 1 – Global Revenue & Customer Insights

This dashboard brings together the primary customer-facing metrics:

- **Revenue by Country** on the left sets the context for which country generate how much revenue in total for that year.
- **Age Group Revenue** and **Gender Contribution** charts follow to show demographic influence.
- The **Revenue by Product Tree map** completes the view by showing how broad or narrow each category's product assortment is.

Two global filters — **Category** and **Customer Gender** — affect every sheet.
Users can click bars, slices, or density cells to filter the rest of the dashboard. All charts update immediately, allowing the user to move from a broad revenue view down into a specific customer or product segment.

Dashboard 2 – Product Performance & Risk Intelligence

This dashboard focuses on product-level behavior, pricing differences, return risk, customer interest, and shipping operations:

- **Shipping Method × Product Heatmap** and **Product Portfolio Overview** form the top row, giving a quick scan of catalog size and whether they rely on expensive shipping methods.
- The second row combines **Final Price Ranking (Lollipop)** with **Discount Profile**, making pricing behavior and markdown patterns easy to compare.
- The bottom row pairs **Popularity Bubbles** with the **Return Severity by Product** , showing which products customers want most and product risk.

Three interaction actions were applied:

- **Product Drilldown Filter** allows a product selection in any sheet to filter the rest of the dashboard.
- **Popularity Bubble Filter** updates pricing, returns, and shipping visuals based on the bubble selected.
- **Shipping Heatmap Filter** reveals product-specific return risk and pricing when a shipping cell is selected.

These interactions allow the user to study each product from multiple angles - risk, demand, pricing, and logistics , without leaving the dashboard.

5. Results

5.1 Global Revenue & Customer Insights (Dashboard 1)

- Books and Electronics emerged as top revenue categories.
- Premium price bands generated a disproportionately high share of revenue.
- Age groups **18 –254**and **35–44** drove the highest revenue.
- Gender split showed a near-even contribution.
- Product type density table revealed an assortment-heavy mix in Electronics & Apparel.

5.2 Product Performance & Risk Intelligence (Dashboard 2)

- Several electronics-related product types showed significantly higher return severity.
- Final price distribution clearly separated premium vs budget product segments.
- Discount profiles showed concentration of promotions around Beauty & Apparel items.
- Popularity bubbles indicated strong demand for sports and electronics products.
- Shipping heatmap exposed operational inefficiencies for certain high-cost product types.

6. Discussion

The two dashboards collectively form a complete intelligence system:

- **Dashboard 1** helps understand *who buys what*, *where*, and *how much they contribute* to revenue.
- **Dashboard 2** helps understand *whether those products are profitable, risky, or operationally expensive*.

Key managerial implications:

- High-return products must undergo quality review or supplier realignment.
- High-popularity, low-price items offer cross-sell opportunities.
- Premium products dominate revenue but may strain express shipping channels.
- Discount-heavy products indicate competitive categories requiring pricing optimization.

The interactive nature of the dashboards allows business users to drill down into individual product types, making insights accessible and actionable.

7. Challenges and Resolutions

- **Flat Distributions:** Some measures (e.g., return rate) had limited variance; solved by using severity-based segmentation.
- **Overlapping Bars:** Solved by adopting a lollipop chart for price ranking.
- **Crowded Bubble Chart:** Reduced clutter via a “Top 20 Popular Products” filter.
- **Shipping Pattern Complexity:** Simplified through a clear product–shipping heatmap.

8. Future Work

- Integrate time-series analysis (orders over months).

- Build a profitability model using cost of goods sold (COGS).
- Add clustering (K-Means) based on price, returns, and popularity.
- Expand dataset with customer lifetime value (CLV) features.

9. References

[1] M. A. Sugraj, “E-Commerce Dataset,” Kaggle, 2023. [Online].
Available: https://www.kaggle.com/datasets/malaiarasugraj/e-commerce-dataset
[2] “E-Commerce Data,” Kaggle. [Online].
Available: https://www.kaggle.com/datasets/carrie1/ecommerce-data
[3] S. Akter and S. F. Wamba, “Big data analytics in e-commerce: a review and framework,” *Electronic Markets*, vol. 26, no. 2, pp. 1–12, 2016.
[4] M. P. Few, “Dashboard Design for Operational Insight,” *Information Dashboard Design*, Analytics Press, 2013.